FinAuth Liveness Detection SDK V5.8.15

Change and Revision History

Version No.	Update Date	Revisions
5.8.0	2025-02-24	Shift of the entity
5.8.1	2025-02-28	SDK active liveness detection supports speed mode and optimized Dynamic liveness
5.8.5	2025-05-30	SDK optimizes the UI configuration, and Android supports 16 KB page size
5.8.6	2025-06-27	SDK adds VoiceOver support and fixes incomplete error prompts during language switching. Dynamic Liveness supports local text customization with improved prompt parameters.
5.8.7	2025-07-04	Recommendation for Update: The SDK should differentiate between optional functions and required functions based on the latest requirements for personal information collection. Please use functions and obtain user authorization according to your actual business needs.
5.8.8	2025-08-06	SDK supports multiple actions during the dynamic liveness action phase. New configuration supports font adjustment based on the phone's system font size.New configuration for distance liveness supports adjustment of long text display position
5.8.9	2025-08-29	SDK supports high-resolution image acquisition functionality
5.8.10	2025-09-19	SDK camera signature 2.0, enhanced anti-injection capabilities
5.8.12	2025-11-19	Improved SDK pass rate, optimized user experience on low-end devices
5.8.13	2026-01-08	Upgrade device risk detection capabilities and enhance SDK client-side security
5.8.14	2026-03-06	SDK supports downgrade configuration for low-end devices and weak network environments, optimizing the on-device success rate through adaptive strategies.
5.8.15	2026-03-18	

		SDK adds dynamic device information collection feature, optimizes device risk detection capability

Integration Diff

You can skip this section for the initial integration. For SDK upgrades, please focus on the change logs between the current version and the target version to avoid upgrade failures. This section has been maintained since version 5.8.12.

Android

Version	aar	model	resource	Proguard rules
5.8.15	update	-	-	-
5.8.14	update	-	-	-
5.8.13	update	-	-	-
5.8.12	update	update	update	update -keep class com.face.lv5.** { *; } -keep class com.zenith.** { *; } -keep class com.neutralbase.safex.** { *; }

iOS

版本	SDK	资源
5.8.15	Need update FinAuthBaseKit.framework FinAuthLiveV5Detect.framework Link Binary With Libraries Need add CoreTelephony.framework libz.tdb The `extraDict` parameter in the initialization interface has been enhanced with additional features. For	Need update FinAuthLiveV5Detect.bundle

	details, please refer to the interface description in section 3.2.1.1.	
5.8.14	Need update FinAuthBaseKit.framework FinAuthLiveV5Detect.framework	Need update FinAuthLiveV5Detect.bundle

1 Introduction to Liveness Authentication

1.1 FaceLiveStill SDK

This SDK is a new version of the liveness authentication SDK provided by Yuanli Jinzhi. A variety of liveness modes are supported by one SDK, in order to make the product safe to provide a perfect experience. Meanwhile, it is characterized by convenient interfacing, compact size, and ease of integration and application, and it can serve various small, medium and large-sized Internet enterprises.

1.2 SDK Compatibility

FaceLiveSDK supports Android, iOS and HarmonyOS platforms.

1. For Android platform,it supports 5.0~16.0;
2. For iOS platform,it supports iOS9 - iOS26 ;
3. For HarmonyOS platform,it supports 5.0.0(12)-6.0.0(20)

In the process of debugging of Android 9 or above devices, a system popup window will appear when you first install it, so as to indicate the use of non-SDK interface (this window only pops up once in the course of debugging). However, this is not the case in release mode.

1.3 Liveness authentication

The logics of liveness authentication roughly involve five steps:

1. After obtaining the api_key and api_secrect through the Yuanli Jinzhi background, perform the SDK authentication to obtain the sign. For details, please refer to the authentication document;
2. Call the Get_biz_token API after obtaining the authentication sign to obtain the biz_token (business serial number);
3. Use biz_token to complete the SDK preprocessing. After the preprocessing is passed, enter the authentication page;
4. After the user passes the authentication process, return to the User Interface and inform the result (error_code, error_message) through the callback;

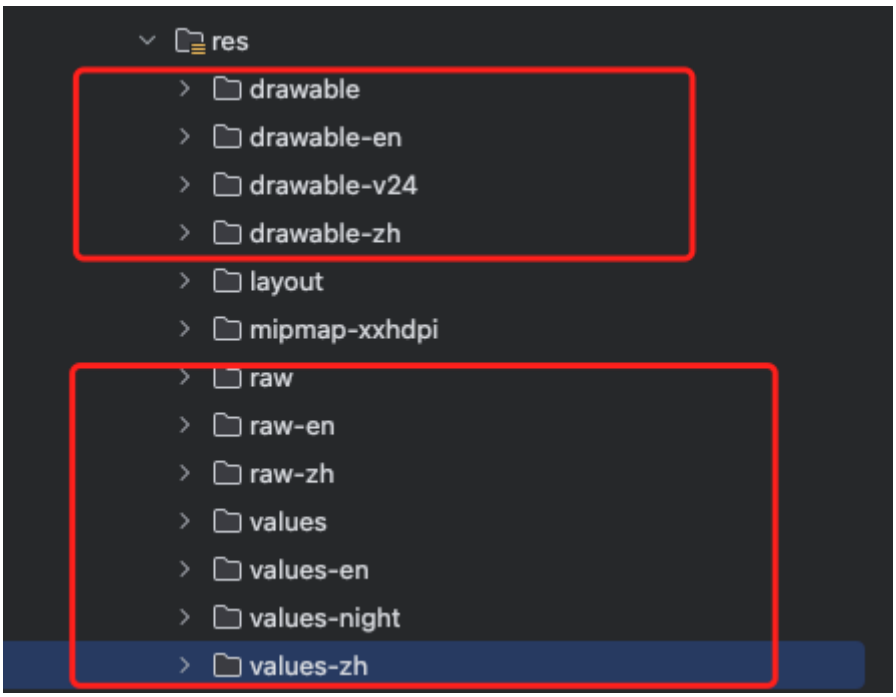
5. Use biz_token to call the Verify API to obtain the result of the current liveness comparison. For details, refer to the Verify documentation.

2 SDK Integration

2.1 Android Integration

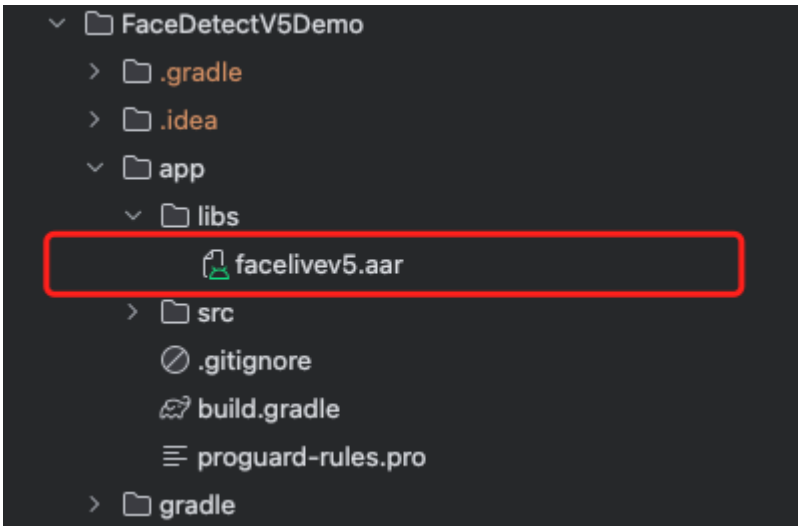
2.1.1 Resource Copy

The resource file in the res directory is copied to res of the program app. If multilingual adaption is not required, you can delete the Chinese and English adaptation folders under res. If adaption to other languages is required,, you need to create a directory in corresponding languages (for image resources, it is also necessary to create a corresponding directory if differentiation between different languages is required). The demo provides three examples: default (Chinese), English, and Chinese



2.1.2 Code Package Copy

Copy the aar file into the libs folder



2.1.3 Gradle Configuration

Add flatDir to the build.gradle of the APP, and add implementation(name:'facelivev5', ext:'aar') to dependencies

```
    repositories{
        flatDir{
            dirs 'libs'
        }
    }
}

dependencies {
    implementation(name:'facelivev5', ext:'aar')
    implementation 'com.google.android.material:material:1.4.0'
```

2.1.4 Screen Recording Function Description

1. There are two ways to apply for screen recording permission. One is to apply at the App layer and set it through the FaceLiveDetectConfig.setMediaProjection() method. The other is that the App does not apply and the SDK layer applies. It is recommended to use the second method. In addition, if the App applies for screen recording permission, the App is responsible for releasing it, otherwise uncontrollable problems will occur.
2. If targetSdkVersion is higher than 28, you need to add it to AndroidManifest.xml

```
<service
  android:name="com.face.lv5.sdk.screen.service.MediaProjectionService"
  android:foregroundServiceType="mediaProjection"
  />
```

```
android {
    compileSdkVersion 29

    defaultConfig {
        applicationId "com.face.demo"
        minSdkVersion 19
        targetSdkVersion 29
        versionCode 1
        versionName "1.0"
```

```
        android:enabled="true"
        />
    <service
        android:name="com.face.lv5.sdk.screen.service.MediaProjectionService"
        android:foregroundServiceType="mediaProjection"
        />
```

2.1.5 Add ProGuard rules

-keep class com.face.lv5.** { *; }

-keep class com.zenith.** { *; }

-keep class com.neutralbase.safex.** { *; }

If NFC is supported, it needs to be added:

-keep class com.eidlink.** {*;}

-keep class com.froad.** { *; }

-keep class com.cn.froad.** { *; }

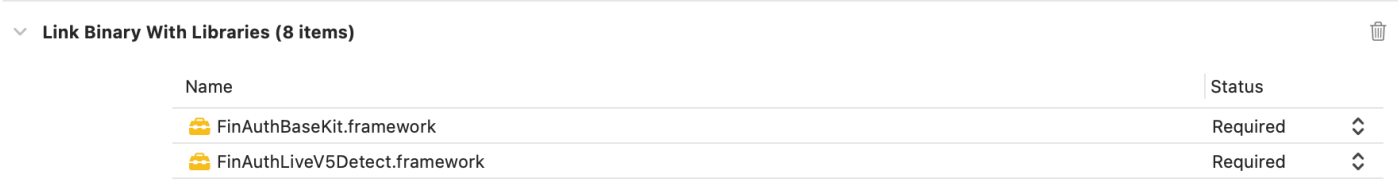
-keep class com.zkteco.** { *; }

2.2 iOS Integration

2.2.1 Add the Framework file to the project

In **Xcode** tool, select *TARGETS -> Build Phases -> Link Binary With Libaries* , click **+** button, click in the pop-up window **Add Other** button, select **FinAuthBaseKit.framework** and **FinAuthLiveV5Detect.framework** are added to the project at the same time.

If you are using the *XCFramework* format file, you can choose **FinAuthBaseKit.framework** and **FinAuthLiveV5Detect.framework** .



2.2.2 Configure Objective-C++ Support

FaceLiveSDK is a static library output. Since it is implemented in *Objective-C++* , you need to ensure that there is at least one source file with the.mm suffix in your project (you can rename any file with the.m suffix to.mm), or specify the compilation method in the project properties, that is, in **Xcode** 's *TARGETS -> Build Setting* , find *Compile Sources As* and set it to *Objective-C++* .

2.2.3 Add resource files to the project

Select the project name, and choose *Add Files to* “Project Name” … in the right-click menu, and select the **FinAuthLiveV5Detect.bundle** file. If you drag the file directly into the project, please pay attention to the directory structure of the file, or check the *Copy items if needed* checkbox, and check *Add to targets* to the project name, and click the **Add** button to add the resource file to the project. For other ways to load resources, please refer to the description of 4.2.3 iOS external resource loading in this document.

2.2.4 Import the required system libraries

In **Xcode** tool, select *TARGETS -> Build Phases -> Link Binary With Libaries* , click **+** button, **Xcode** project needs to introduce **AVFoundation.framework** , **CoreMedia.framework** , **CoreMotion.framework** , **SystemConfiguration.framework** , **WebKit.framework** , **MediaPlayer.framework** , **ReplayKit.framework** , **CoreTelephony.framework** , **libz.tbd** system library.

Link Binary With Libraries (9 items)

Name	Status
AVFoundation.framework	Required
CoreMedia.framework	Required
CoreMotion.framework	Required
SystemConfiguration.framework	Required
WebKit.framework	Required
MediaPlayer.framework	Required
ReplayKit.framework	Required
CoreTelephony.framework	Required
libz.tbd	Required

Drag to reorder linked binaries

2.2.5 Add project permissions

Since FaceLiveSDK needs to use the device camera, please add **Privacy - Camera Usage Description** permission to the *info.plist* file.

FaceIDLiveDetect 〉 FaceIDLiveDetect 〉 Info 〉 No Selection

Key	Type	Value
Information Property List	Dictionary	(16 items)
Default localization	String	$(DEVELOPMENT_LANGUAGE)
Bundle display name	String	V5集成示例Demo
Executable file	String	$(EXECUTABLE_NAME)
Bundle identifier	String	$(PRODUCT_BUNDLE_IDENTIFIER)
InfoDictionary version	String	6.0
Bundle name	String	$(PRODUCT_NAME)
Bundle OS Type code	String	APPL
Bundle version string (short)	String	1.0
Bundle version	String	1
Application requires iPhone environment	Boolean	YES
Privacy - Camera Usage Description	String	App需要您的同意才能访问相机
Launch screen interface file base name	String	LaunchScreen
Main storyboard file base name	String	Main
Required device capabilities	Array	(1 item)
Supported interface orientations	Array	(3 items)
Supported interface orientations (iPad)	Array	(4 items)

2.3 HarmonyOS Integration

2.3.1 SDK Version and Support Instructions

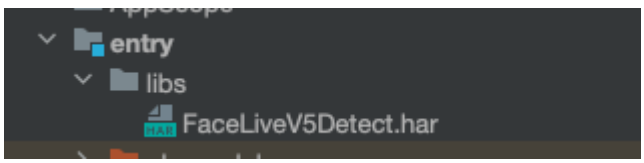
The SDK supports a minimum system of 5.0.0(12)

2.3.2 SDK Package Directory Description

1. demo
2. sdk

2.3.3 Integration Steps

- Put FaceLiveV5Detect.har under the project directory, as shown in the following figure:



- Configure the oh - package.json5 in the current Model, associate the SDK in dependencies, the path is the actual path of FaceLiveV5Detect.har, and the name is specified as facelivev5detect, as shown in the following figure:

```
{
  "license": "",
  "devDependencies": {},
  "author": "",
  "name": "entry",
  "description": "Please describe the basic information.",
  "main": "",
  "version": "1.0.0",
  "dynamicDependencies": {},
  "dependencies": {
    "facelivev5detect": "file:./libs/FaceLiveV5Detect.har"
  }
}
```

3 SDK Interfaces

3.1 Android Interfaces

3.1.1 Entry class FaceLiveManager

The FaceLiveManager class is a class that manages liveness authentication. This class cannot be initialized and can only obtain its singleton through getInstance.

3.1.1.1 Get the singleton object of FaceLiveManager

Function name	getInstance
FaceLiveManager getInstance()	
Description	Get an instance of FaceLiveManager
Type description	function
Remarks	Get the singleton of FaceLiveManager
Return value	
FaceLiveManager	An example

3.1.1.2 Start detection

Function name	startDetect

void startDetect(Context context, FaceLiveDetectConfig config, FaceLiveDetectListener callback)		
Description	Start liveness detection	
Type description	function	
Remarks	Enable liveness detection interface.	
Variable	**Remarks**	**Type**
context	Android context	Context
config	For parameter configuration, see FaceLiveDetectConfig for details	FaceLiveDetectConfig
callback	The implementation class of the FaceLiveDetectListener interface, which accepts the interface of the liveness detection result, see the callback function FaceLiveDetectListener for details	FaceLiveDetectListener

3.1.1.3 Set callback of the data synchronization interface in transparent mode

Function name	setDataSyncCallback	
void setDataSyncCallback(FaceLiveDataSyncCallback callback)		
Description	Transparent transmission data synchronization callback	
Type description	function	
Remarks	Receive data that needs to be synchronized through this callback interface in transparent transmission mode	
Variable name	**Remarks**	**Type**
callback	The implementation class of the FaceLiveDataSyncCallback interface receives the types and data that need to be synchronized. See the callback function FaceLiveDataSyncCallback for details.	FaceLiveDataSyncCallback

3.1.1.4 Retrieve log information during the liveness authentication process

Function name	getSDKLog

String getSDKLog()	
Description	Retrieve the log information during the liveness authentication process
Type description	function
Remarks	None
Return value	
String	Encrypt the log information during the liveness authentication process

3.1.1.5 Decryption of video and image files in the course of liveness detection

Function name	startDetectForLivenessFile	
LivenessFileResult getLivenessFiles(String filePath, String decryptionKey)		
Description	Get and decrypt liveness files	
Type description	function	
Remarks		
Return value	LivenessFileResult	
Variable name	**Remarks**	**Type**
filePath	The path of encrypted liveness file returned by the SDK after successful terminal liveness detection	String
decryptionKey	Decryption key obtained after call of verify	String

3.1.1.6 Get SDK version information

Function name	getVersion
String getVersion()	
Description	Get the SDK version
Type description	function
Remarks	None
Return value	

String	Current SDK version

3.1.1.7 Getting SDK build information

Function name	getBuildInfo
String getBuildInfo()	
Description	Get SDK build information
Type description	function
Remarks	None
Return value	
String	A string

3.1.1.8 Setting callback method for compatibility mode

Function name	setImageDataListener	
void setImageDataListener(FaceLiveImageDataListener listener)		
Description	Setting Callback Method	
Type description	function	
Remarks	None	
Variable name	**Remarks**	**Type**
listener	Callback function for compatibility mode, see FaceLiveImageDataListener class for details	FaceLiveImageDataListener

3.1.2 FaceLiveDetectConfig detection configuration class

Variable	Type	Description
bizToken	String	Requesting token, mandatory parameter.
modelPath	String	Model path
host	String	Service host

language	String	SDK language. Please provide the parameters according to the ISO 639 language codes standard.
isShowLogo	boolean	Whether to display the bottom logo
autoAdjustVolume	boolean	Automatic adjustment of audio prompt volume for active liveness, compatible mode (optional), default: false.
suggestVolume	int	Recommended volume adjustment value, compatible mode (optional), default value: 50, when autoAdjustVolume is set to true
verticalDetection	int	Vertical detection: Continuously detect. 1. Detect for the first two seconds. 2. Do not detect.
mediaProjection	MediaProjection	Object after applying for screen recording permission
isEnterLoadingPage	boolean	Whether to enter the Loading page
isLandscape	boolean	Whether the pad is horizontally screened (mobile phone horizontal screen is not supported)
mode	int	0: Normal Mode 3: Compatibility Mode 5: Transparent Mode
customTimeout	int	Interface waiting timeout in pass-through mode, default is 30s
configData	String	Liveness configuration information. • When `mode` is equal to 3, `configData` is a required field.
hostList	List<String>	Backup Host List: • The backup host list is valid when `mode` is equal to 3. • If the request to the primary host fails, the addresses in the `hostList` will be attempted.
isMute	boolean	Whether to mute
isHiddenActionTime	boolean	Whether to hide the countdown of the action in the Dynamic liveness action stage true: Hide false: Show(default)

fontFollow	boolean	The SDK font will adaptively adjust its size according to the mobile phone system font. false: Off (default) true: On
distanceTipsDown	boolean	Whether the prompt for living body detection is displayed below the face frame only takes effect in long mode (multi-line mode). false: Off (default) true: On
livenessVideoSeparate	boolean	Should local retention split into two files: image and video? false: Do not split (default) true: Split
config_max_time	int	Default: 10s
uploadTimeout	int	Default: 5s

3.1.3 FaceLiveDetectListener detection listening callback

3.1.3.1 onPreDetectFinish

Function name	onPreDetectFinish	
void onPreDetectFinish(int errorCode, String errorMessage);		
Description	Callback of pre-processing result	
Type description	function	
Remarks		
Return value		
Variable name	**Remarks**	**Type**
errorCode	See 3.1.4 for error codes	int
errorMessage	Error description	String

3.1.3.2 onDetectFinish

Function name	onDetectFinish

void onDetectFinish(int errorCode, String errorMessage,String bizToken)		
Description	Callback for detection results	
Type description	function	
Remarks		
Return value		
Variable name	**Remarks**	**Type**
errorCode	See 3.1.4 for error codes	int
errorMessage	Error description	String
bizToken	Business No.	String

3.1.3.3 onLivenessFileCallback

Function name	onLivenessFileCallback	
void onLivenessFileCallback(String livenessFilePath)		
Description	Callback of liveness data retention result	
Type description	function	
Remarks		
Return value		
Variable name	**Remarks**	**Type**
livenessFilePath	The path of liveness retention file	String

3.1.3.4 onLivenessLocalFileCallBack

Function name	onLivenessLocalFileCallBack
void onLivenessLocalFileCallBack(FaceliveLocalFileInfo fileInfo)	
Description	Callback of liveness data retention result, including liveness retention and screen record retention
Type description	function

Remarks		
Return value		
Variable name	**Remarks**	**Type**
fileInfo	Local retention file class, see faceliveLocalFileInfo class for details	FaceliveLocalFileInfo

3.1.3.5 FaceliveLocalFileInfo

Description	Local retention file class	
Type description	Class	
Remarks		
Return value		
Variable name	**Remarks**	**Type**
filePath	The path of liveness data retention file	String
scrrenFilePath	The path of screen record data retention file	String

3.1.4 FaceLiveImageDataListener Compatibility Mode Detection Result Callback

3.1.4.1 onImageData

Function name	onImageData	
void onImageData(HashMap<String,byte[]> imageData, byte[] delta)		
Description	Callback for compatibility mode detection results	
Type description	function	
Remarks	See (5.5 SDK compatibility Mode Image List) for the key values in imageData	
Return value		
Variable name	**Remarks**	**Type**
bestImage	best image	HashMap
delta	liveness detection result data	byte[]

3.1.5 FaceLiveDataSyncCallbacktransparent mode synchronization interface callback

3.1.5.1 performDataSync

Function name	performDataSync	
void performDataSync(String bizToken,int syncType, String syncData, FaceLiveDataSyncResponse response);		
Description	Data synchronization callback	
Type description	function	
Remarks		
Return value		
Variable name	**Remarks**	**Type**
bizToken	bizToken for current request	String
syncType	Data synchronization type	int
syncData	Data to be synchronized	String
response	Callback for the result after the SDK interface request, see 3.1.6 FaceLiveDataSyncResponse	FaceLiveDataSyncResponse

3.1.5.2 onGetFaceliveData

Function name	onGetFaceliveData	
void onGetFaceliveData (byte[] faceliveData);		
Description	faceliveData data callback	
Type description	function	
Remarks	The SDK returns faceliveData in transparent mode	
Return value		
Variable name	**Remarks**	**Type**

faceliveData	Liveness detection data	byte[]

3.1.6 FaceLiveDataSyncResponse transparent mode notification SDK synchronization result interface

3.1.6.1 onResponse

Function name	onResponse	
void onResponse(int syncType, String response);		
Description	PTransparent mode returns SDK synchronization result interface	
Type description	function	
Remarks	Called after requesting the server level synchronization interface	
Return value		
Variable name	**Remarks**	**Type**
syncType	Data synchronization type	int
response	Data synchronization results	String

3.1.7 Error Code

ErrorCode	ErrorMessage	Remarks
1000	LIVENESS_FINISH	Liveness completion
	GET_CONFIG_SUCCESS	Successfully obtained the configuration
1001	BIZ_TOKEN_DENIED	The incoming biz_token does not meet the requirements
1002	ILLEGAL_PARAMETER:{livenesstype} ILLEGAL_PARAMETER:{Context} ILLEGAL_PARAMETER:{FaceLiveDetectConfig} ILLEGAL_PARAMETER:{FaceLiveDetectConfig:model} ILLEGAL_PARAMETER:{FaceLiveDetectConfig:appKey} ILLEGAL_PARAMETER:{FaceLiveDetectConfig:livenessId}	The imported parameter is illegal, and the specific reason is in {}

	ILLEGAL_PARAMETER:{FaceLiveDetectConfig:bizToken} ILLEGAL_PARAMETER:{FaceLiveDetectConfig:host} ILLEGAL_PARAMETER:{FaceLiveDetectConfig:modelPath} ILLEGAL_PARAMETER:{missing_liveness_config} ILLEGAL_PARAMETER:{response_result_is_null} ILLEGAL_PARAMETER:{request_data_is_null} ILLEGAL_PARAMETER:{request_data_error} ILLEGAL_PARAMETER:{jsonexception_200} ILLEGAL_PARAMETER:{jsonexception_400}	
1003	AUTHENTICATION_FAIL:{illegal_param} AUTHENTICATION_FAIL:{illegal_handle} AUTHENTICATION_FAIL:{illegal_index} AUTHENTICATION_FAIL:{expire} AUTHENTICATION_FAIL:{bundleid_error} AUTHENTICATION_FAIL:{license_error} AUTHENTICATION_FAIL:{liveness_id_error} AUTHENTICATION_FAIL:{model_error} AUTHENTICATION_FAIL:{algo_error} AUTHENTICATION_FAIL:{opengl_context_error}	Authentication failed, the specific reason is in {}
1004	MOBILE_PHONE_NOT_SUPPORT	The phone is not supported
1005	Specific exception information, format: exception type_code class name_code line number Example: NullPointException_com.face.xx.Test_1101	If such an error occurs, it is recommended to contact technical support in time
1006	REQUEST_FREQUENTLY	When the same device is called multiple times at the same time, the first call runs normally, and other calls return this type of error
1007	NETWORK_TIME_OUT	Network request timed out
1008	INTERNAL_ERROR	Network configuration error. When this error occurs, please request again. If this error persists, please contact

		customer service or business in time.
1009	INVALID_BUNDLE_ID	Information verification failed. Please restart the program or device and try again.
1010	NETWORK_ERROR	Unable to connect to the Internet. Please connect to the Internet and try again.
1011	USER_CANCEL	User cancelled
1012	NO_CAMERA_PERMISSION	You don't have permission to open the camera. Please enable the permission and try again.
1013	DEVICE_NOT_SUPPORT	Unable to start the camera. Please make sure the camera is working properly.
1014	FACE_INIT_FAIL:{model init failed} FACE_INIT_FAIL:{init data failed} FACE_INIT_FAIL:{surfaceTexture is null} FACE_INIT_FAIL:{errorCode}	Unable to start facial recognition, please try again later
1016	LIVENESS_FAILURE	liveness detection failed
1017	GO_TO_BACKGROUND	The app goes to the background and the liveness detection fails
1018	LIVENESS_TIME_OUT	Operation timed out because the user did not perform any operations for a long time
1019	DATA_UPLOAD_FAIL	Data upload failed
1022	MOBILE_PHONE_NOT_SUPPORT_SCRN	The device does not support screen recording
1023	SCRN_AUTHORIZATION_FAIL	Screen recording authorization failed
1024	SCRN_RECORD_FAIL	Video recording failed
1025	VIDEO_SAVE_FAIL	Video save failed

1026	NO_AUDIO_RECORD_PERMISSION	No permission to record voice
4205	*LOAD_LIB_FAILED*	Load library failed

3.2 iOS Interface Description

3.2.1 FinAuthLiveV5DetectManager class

The FinAuthLiveV5DetectManager class is the core class for managing liveness detection. All functions related to liveness detection need to use this class.

3.2.1.1 Initialize liveness detection authentication

Function name	+faceLiveDetectManagerWithBizToken: configInfo:extraData:startCallBack:detectVC:endCallBack:dismissCallBack:
+ (void)faceLiveDetectManagerWithBizToken:(NSString *__nonnull)bizTokenStr configInfo:(FinAuthLiveV5DetectInitConfigItem *__nullable)configItem extraData:(NSDictionary *__nullable)extraDict startCallBack:(FinAuthLiveV5StartDetectBlock)startBlock detectVC:(UIViewController *)detectVC endCallBack:(FinAuthLiveV5EndDetectBlock)endBlock dismissCallBack:(FinAuthLiveV5DismissBlock)dismissBlock;	
Remarks	1. Start FinAuth liveness detection, support integration mode FinAuthLiveV5DetectModel_ *A and FinAuthLiveV5DetectModel* _C. 2. Set different liveness detection types to obtain `bizTokenStr`, and return the verification result through `block` after the SDK is started. 3. Different blocks are returned at different stages. If the content in `startBlock` has prompted a failure, the block for the next stage will not be returned.

Parameter	**Remarks**	**Type**
bizTokenStr	Set the startup configuration for FinAuth liveness detection. Please use the get_biz_token related interface to obtain this parameter	NSString *__nonnull
configItem	Set the custom configuration for FinAuth liveness detection. For details, please refer to the description of the	FinAuthLiveV5DetectInitConfigItem *__nullable

	`FinAuthLiveV5DetectInitConfigItem` class.	
extraDict	A reserved parameter. **Currently supported configurations:** 1. **datasync_max_time** Type: `NSNumber (int)` Default: `30` Range: `[1, 120]` Description: Maximum waiting time in passthrough mode. Example: `@{@"datasync_max_time": @(30)}` 2. **config_max_time** Type: `NSNumber (int)` Default: `10` Range: `[1, 120]` Description: Timeout duration for fetching configuration requests. Example: `@{@"config_max_time": @(30)}` 3. **config_re_request** Type: `NSNumber (int)` Default: `2` Range: `[1, 10]` Description: Number of retry attempts for fetching configuration requests. Example: `@{@"config_re_request": @(3)}` 4. **upload_timeout_time** Type: `NSNumber (int)` Default: `5` Range: `[1, 60]` Description: Timeout duration for uploading detection results. Example: `@{@"upload_timeout_time": @(30)}`	NSDictionary *__nullable

	5. **DisAllowDetectRootVC** Type: `NSNumber (BOOL)` Default: `false` Description: Whether to disable top-level view controller detection. When set to `true`, top-level view detection will be skipped, and initialization will no longer return the error: `ILLEGAL_PARAMETER: {detectVC}` 6. **ShowV5Loading** Type: `NSNumber (BOOL)` Default: `false` Description: Whether to display the loading page. When set to `true`, a loading screen will be shown before the detection page appears.	
startBlock	block at the time of completion of liveness detection initiation	FinAuthLiveV5StartDetectBlock
detectVC	A VC page used to enable liveness detection, which is generally the current ViewController	UIViewController *
endBlock	block at the time of completion of liveness detection	FinAuthLiveV5EndDetectBlock
dismissBlock	block at the time of the end of liveness detection and the closing of pages	FinAuthLiveV5DismissBlock
Return value	No return value	

3.2.1.2 Initialize liveness detection authentication

Function name	+faceLiveDetectManagerWithBizToken: configInfo:extraData:networkCallBack:startCallBack:detectVC:endCallBack:dismissCallBack:

```
+ (void)faceLiveDetectManagerWithBizToken:(NSString *__nonnull)bizTokenStr
 configInfo:(FinAuthLiveV5DetectInitConfigItem *__nullable)configItem
 extraData:(NSDictionary *__nullable)extraDict
```

```
networkCallBack:(FinAuthLiveV5NetworkRequestBlock)networkBlock
startCallBack:(FinAuthLiveV5StartDetectBlock)startBlock
detectVC:(UIViewController *)detectVC
endCallBack:(FinAuthLiveV5EndDetectBlock)endBlock
dismissCallBack:(FinAuthLiveV5DismissBlock)dismissBlock;
```

Remarks	1. Turn on the FinAuth liveness detection, support the integration methods of FinAuthLiveV5DetectModel_A, FinAuthLiveV5DetectModel_C and FinAuthLiveV5DetectModel_E. 2. Set different liveness detection types to obtain `bizTokenStr`, and return the verification result through `block` after the SDK is started. 3. Different blocks are returned at different stages. If the content in `startBlock` has prompted a failure, the block for the next stage will not be returned.

Parameter	Remarks	Type
bizTokenStr	Set the startup configuration for FinAuth liveness detection. Please use the get_biz_token related interface to obtain this parameter	NSString *__nonnull
configItem	Set the custom configuration for FinAuth liveness detection. For details, please refer to the description of the `FinAuthLiveV5DetectInitConfigItem` class.	FinAuthLiveV5DetectInitConfigItem *__nullable
extraDict	Reserved parameter. Same as the parameter in section 3.2.1.1.	NSDictionary *__nullable
networkBlock	Network request block. When configItem.model is FinAuthLiveV5DetectModel_E, this parameter is required. This block will return multiple times. Please perform operations according to the networkType.	FinAuthLiveV5NetworkRequestBlock
startBlock	block at the time of completion of liveness detection initiation	FinAuthLiveV5StartDetectBlock
detectVC	A VC page used to enable liveness detection, which is generally the current ViewController	UIViewController *
endBlock	block at the time of completion of liveness detection	FinAuthLiveV5EndDetectBlock
dismissBlock		FinAuthLiveV5DismissBlock

	block at the time of the end of liveness detection and the closing of pages	
Return value	No return value	

3.2.1.3 Get the log information during the liveness authentication

Function name	+queryFinAuthLiveDetectLogInfo
+ (NSData *)queryFinAuthLiveDetectLogInfo;	
Remarks	The log information is encrypted data. Please decrypt it through Yuanli Service.
Parameter	No parameters
Return value	**Return value description**
NSData *	Encrypted log data

3.2.1.4 Decrypt video and image files during liveness detection

Function name	+encodeFinAuthLiveDetectFileWithFilePath:encodeKey:	
+ (NSDictionary *)encodeFinAuthLiveDetectFileWithFilePath:(NSString *)filePath encodeKey:(NSString *)encodeStr;		
Remarks	Decryption processing is performed on the locally retained data returned by the SDK, and the user is required to provide a `key` for decryption. The `key` can be obtained through the verify interface.	
Parameter	**Remarks**	**Type**
filePath	Path to store encrypted files	NSString *
encodeStr	Key for decryption	NSString *
Return value	**Return value description**	
NSDictionary *	Decrypted file information. This information is a combination of the file name and path temporarily stored in the sandbox after decryption.	

3.2.1.5 Reset the FinAuth request queue

Function name	+resetFinAuthFrequentlyRequest
+ (void)resetFinAuthFrequentlyRequest;	
Remarks	If the error `FinAuthLiveV5DetectErrorTypeRequestFrequently` is always prompted when calling `+faceLiveDetectManagerWithBizToken:`, but the method of starting FinAuth liveness detection is not called frequently, please use this method to reset the FinAuth request queue.
Parameter	No parameters
Return value	No return value

3.2.1.6 Get SDK version information

Function name	+getSDKVersion
+ (NSString *_Nonnull)getSDKVersion	
Remarks	None
Parameter	No parameters
Return value	Return value description
NSString *	SDK version number

3.2.1.7 Get SDK build information

Function name	+getSDKBuild
+ (NSString *_Nonnull)getSDKBuild	
Remarks	None
Parameter	No parameters
Return value	Return value description
NSString *	SDK build version information

3.2.2 Configurable Classes

This type of file includes the configuration that the user can customize.

3.2.2.1 Class FinAuthLiveV5DetectInitConfigItem

Initialize the configuration used by the liveness detection SDK

Variable name	Remarks	Default value	Type
languageType	It is mandatory. Specify the liveness V5 language type. For details on how to load SDK language resources, please refer to the resource description in the document	FinAuthLiveV5DetectLanguageTypeCh Chinese	FinAuthLiveV5DetectLanguageType
model	It is mandatory. Specify the integration method of FinAuthV5SDK.	FinAuthLiveV5DetectModel_A Default integration method	FinAuthLiveV5DetectModel
hostURL	Non - essential Specify the HOST address for the FinAuthV5SDK to pull the configuration.	https://api.yljz.com Used in Chinese mainland	NSString *
configInfo	Conditions must This parameter is required when the model is `FinAuthLiveV5DetectModel_C`. If the model is set to `FinAuthLiveV5DetectModel_C`, but this parameter is nil, the SDK initialization will fail. This parameter is invalid for other model types. It needs to be obtained through the get_biz_token interface.	nil	NSString *
standbyURLList	Optional. List of alternative URLs for authorization requests.	nil	NSArray<NSString *>*

	When the request using the `hostURL` parameter fails, use the URLs in this list to retry. This parameter is valid when the model is `FinAuthLiveV5DetectModel_C`. This parameter is invalid for other model types.		
customUI	Optional. Customize the UI style of the FinAuthV5SDK	nil	FinAuthLiveV5DetectUIConfigItem*
bundleFilePath	Optional. Specify the absolute path of the SDK resource, ending with **bundle**. If the value is `nil` or `""`, the resource is read from the MainBundle. For details on how to load resources, see the resource description in the documentation.	nil	NSString *
phoneVertical	Optional. Specify the vertical detection type of the device during the liveness detection process.	FinAuthLiveV5DetectPhoneVerticalTypeDisable	FinAuthLiveV5DetectPhoneVerticalType
isAdjustPhoneVolume	Optional. Whether to adjust the volume. YES is on and NO is off. After being turned on, the current device volume will be adjusted to 'maxPhoneVolume' (NO by default)	NO	BOOL
adjustPhoneVolume	Optional.	0	int

	Maximum volume after volume adjustment. The threshold range is [0, 100], and the default is 0. This parameter only takes effect when the 'isAdjustPhoneVolume' value is YES If the setting value is less than the current system volume, the setting is not performed.		
isMute	Optional. Set to mute mode. YES means setting the mute mode, NO means not setting it. The default value is NO	NO	BOOL
showPoweryby	Optional. YES means display, NO means not display. Set whether to display the powerby image at the bottom of the liveness detection page.	NO	BOOL
isHiddenAction Time	Optional. YES means hidden, NO means not hidden. Specify whether to hide the countdown prompt for the action stage of the Dynamic liveness detection	NO	BOOL

3.2.2.2 FinAuthLiveV5DetectUIConfigItem

Customize the configuration used by the liveness detection SDK UI

Variable name	Remarks	Default value	Type

followsSystemFontSize	Whether to follow the system font size. The default value is NO, do not follow. After setting it to YES, follow the system font size change. When set to YES, other custom font size configurations in this class, such as livenessHomeExitPopupwindowTextSize, no longer take effect. When set to YES, all text in the live page uses the system font size, if the device uses the oversized font size in the accessibility function, it may cause the UI to display abnormally, please turn it on with caution.	NO	BOOL
remindUIType	Prompt UI effect type	FinAuthLiveV5DetectRemindUITypeDefault	FinAuthLiveV5DetectRemindUIType
multipleLineDistanceRemindLocation	Distanse live position of the live prompt text under the multipleLine effect. NO is shown at the top, YES is shown at the bottom. Effective only when remindUIType==FinAuthLiveV5DetectRemindUITypeMultipleLine	NO	BOOL
iPadAuthRotate	To support iPad devices such as the 6th-generation iPad Air, whose camera is positioned for landscape use, this SDK follows the app’s interface rotation settings by default. • If the app supports portrait mode only, the liveness detection page will support portrait mode only. • If the app supports landscape mode only, the liveness detection page will support landscape mode only. • If the app supports both portrait and landscape rotation, the liveness detection page will support both orientations.	0	int

	For more advanced customization, please configure the `iPadAuthRotate` property. • `0` – Follow the system (default) • `1` – Disable rotation and force portrait mode • `2` – Disable rotation and force landscape mode Default value: `0`. **Note:** • If the app does not support landscape mode, do not set this property to `2`, as it may cause the app to crash. • If the app does not support portrait mode, do not set this property to `1`, as it may cause the app to crash.		
livenessHomeBackgroundColor	Background color of non - face area	0xFFFFFF	UIColor *
livenessHomeNormalRemindTextColor	The color of the prompt text when entering the mirror stage for the first time	0x292929	UIColor *
livenessHomeFailedRemindTextColor	Text color of the prompt for non-first entry into the mirror stage	0xFF3B42	UIColor *
livenessHomeDeviceVerticalRemindColor	Vertical vertical prompt font color of mobile phone Only valid when remindUIType == FinAuthLiveV5DetectRemindUITypeDefault	0xFFFFFF	UIColor *
livenessHomeMultipleLineDeviceVerticalRemindColor	Vertical vertical prompt font color on mobile phone under multi - line effect Only valid when remindUIType == FinAuthLiveV5DetectRemindUITypeMultipleLine	0x292929	UIColor *
livenessHomeProcessBarColor	Dynamic liveness Progress Bar Color Only the active liveness mode and the flash liveness without light mode are effective, while the flash liveness with light mode is not effective	0x267CE0	UIColor *

livenessHomeActionHatColor	The top shadow color during the active liveness process	0x0678FC	UIColor *
livenessHomeCheckingLineStartColor	Verification phase Progress Bar color, gradual change	0x267CE0	UIColor *
livenessHomeCheckingLineEndColor	Verification phase Progress Bar color, gradual change	0xFFFFFF	UIColor *
livenessHomeExitPopupwindowTextSize	Exit Popup Title Font Size	17	CGFloat
livenessHomeExitPopupwindowBodySize	Exit pop - up body font size	13	CGFloat
livenessHomeConfirmButtonColor	Exit the color of the confirmation button in the pop - up box	0x000000	UIColor *
livenessHomeCancelButtonColor	Exit the color of the cancel button in the pop - up box	0x000000	UIColor *
livenessHomeAgreementpageTitleTextSize	Font size of the top title on the agreement page	26	CGFloat
livenessHomeAgreementpageBottomTitleTextSize	Font size of the prompt at the bottom of the agreement page	14	CGFloat
livenessHomeAgreementpageBottomButtonBeforeClickColor	Normal state color of the button on the agreement page	0x3B94FC	UIColor *
livenessHomeAgreementpageBottomButtonAfterClickColor	Highlight attitude color of protocol page button	0x267CE0	UIColor *
livenessHomeCustomPromptBackgroundColor	Liveness interface global prompt background color	0xFDEBEB	UIColor *
livenessHomeCustomPromptTextColor	Global prompt font color for the liveness interface	0xDA3130	UIColor *
livenessCustomPromptColor	Prompt font color for the liveness interface confirmation mode	0xD3463E	UIColor *
livenessHomeCustomPromptBackgroundColor	Liveness interface global prompt background color	0xFDEBEB	UIColor *
livenessHomeCustomPromptTextColor	Global prompt font color for the liveness interface	0xDA3130	UIColor *

livenessCustomPromptColor	Prompt font color for the liveness interface confirmation mode	0xD3463E	UIColor *

3.2.3 FinAuthLiveV5DetectError class

Error message description class.

Variable name	Remarks	Default value	Type
errorCode	liveness authentication return value code	FinAuthLiveV5DetectErrorTypeOK	FinAuthLiveV5DetectErrorType
errorMessage	liveness authentication return value message	nil	NSString *

3.2.4 Enumeration

3.2.4.1 FinAuthLiveV5DetectModel

Integrated method of liveness detection

Enumeration name	Remarks	Enumeration value
FinAuthLiveV5DetectModel_A	Default integration method Authorization and lliveness configuration acquisition through network requests within the SDK need to be used in conjunction with the specified'hostURL 'parameter	1
FinAuthLiveV5DetectModel_C	Compatibility integration mode Authorization is only performed through network requests within the SDK. Lliveness configuration requires the user to actively pass in and needs to be used in conjunction with the specified'hostURL 'and'configInfo' parameters	3
FinAuthLiveV5DetectModel_E	Data pass-through integration method No network requests are made within the SDK. The request body data of all network requests is passed to the APP through blocks, and some data needs to be requested by the APP through the network and then returned to the SDK using blocks.	5

3.2.4.2 FinAuthLiveV5DetectLanguageType

Language type for liveness detection

Enumeration name	Remarks	Enumeration value
FinAuthLiveV5DetectLanguageTypeCh	Chinese mode	0

3.2.4.3 FinAuthLiveV5DetectPhoneVerticalType

Vertical detection type of the liveness detection device

Enumeration name	Remarks	Enumeration value
FinAuthLiveV5DetectPhoneVerticalTypeContinue	Continuously enable device vertical detection function	1
FinAuthLiveV5DetectPhoneVerticalTypeFront2	Enable only within 2 seconds of detection activation, then disable this function	2
FinAuthLiveV5DetectPhoneVerticalTypeDisable	Disable device vertical detection function	3

3.2.4.4 FinAuthLiveV5DetectRemindUIType

UI effect type of liveness detection prompt

Type	Remarks	value
FinAuthLiveV5DetectRemindUITypeDefault	One-line mode for hint effect	0
FinAuthLiveV5DetectRemindUITypeMultipleLine	Prompt effect multi-line mode, up to three lines are supported	1

3.2.4.5 FinAuthLiveV5DetectErrorType

Liveness detection error type

Type	Remarks	value
FinAuthLiveV5DetectErrorTypeOK	SDK liveness succeeded	1000
		1001

FinAuthLiveV5DetectErrorTypeBizTokenDenied	Incoming biz_ Token does not meet the requirements	
FinAuthLiveV5DetectErrorTypeIllegalParameter	Incoming parameter is illegal	1002
FinAuthLiveV5DetectErrorTypeAuthenticationFail	SDK authentication failed	1003
FinAuthLiveV5DetectErrorTypeMobileNotSupport	The phone is not on the support list	1004
FinAuthLiveV5DetectErrorTypeNullPointException	If such an error occurs, please contact FinAuth technical support	1005
FinAuthLiveV5DetectErrorTypeRequestFrequently	Multiple calls from the same device simultaneously	1006
FinAuthLiveV5DetectErrorTypeNetworkTimeout	Network request timed out	1007
FinAuthLiveV5DetectErrorTypeInternalError	Network configuration error. When such an error occurs, please try again. If this error persists, please contact the Force Intelligence technical support	1008
FinAuthLiveV5DetectErrorTypeInvalidBundleID	Information verification failed, please try again	1009
FinAuthLiveV5DetectErrorTypeNetworkError	Network connection failed. Please check your network status	1010
FinAuthLiveV5DetectErrorTypeUserCancel	Cancelled by the user	1011
FinAuthLiveV5DetectErrorTypeNoCameraPermission	You do not have permission to use the camera. Please enable camera permission and try again	1012
FinAuthLiveV5DetectErrorTypeNoCameraSupport	The camera cannot be enabled. Please make sure the camera is functional	1013
FinAuthLiveV5DetectErrorTypeFaceInitFail	Face recognition cannot be enabled. Please try again	1014

FinAuthLiveV5DetectErrorTypeLivenessFailure	SDK liveness detection failed	1016
FinAuthLiveV5DetectErrorTypeGotoBackground	The application was pushed to the background and the liveness detection failed	1017
FinAuthLiveV5DetectErrorTypeLivenessTimeout	Application operation timeout. The liveness detection failed	1018
FinAuthLiveV5DetectErrorTypeDataUploadFail	Liveness detection upload is abnormal. Liveness detection failed	1019
FinAuthLiveV5DetectErrorTypeNFCUserQuit	The user exits the NFC detection process	1020
FinAuthLiveV5DetectErrorTypeNFCSystemQuit	The NFC detection process system exits abnormally	1021
FinAuthLiveV5DetectErrorTypeScreenNoSupport	This device does not support screen recording	1022
FinAuthLiveV5DetectErrorTypeScreenNoPermission	Failed to obtain the screen recording permission	1023
FinAuthLiveV5DetectErrorTypeScreenRecordFail	Screen recording failed	1024
FinAuthLiveV5DetectErrorTypeScreenWriteFail	Screen recording and video saving failed	1025
FinAuthLiveV5DetectErrorTypeNoAudioPremission	The device does not have microphone permission. The iOS platform does not return this error. It is only used to unify errors with other platforms.	1026
FinAuthLiveV5DetectErrorTypeVideoRecordFail	Unable to save the file locally, liveness detection failed.	1027

3.2.5 Block

3.2.5.1 FinAuthLiveV5StartDetectBlock

Block Name	FinAuth liveness detection Configuration Interface Manager Loading Status Block

typedef void(^FinAuthLiveV5StartDetectBlock)(FinAuthLiveV5DetectError* error, NSDictionary* extraOutDict);	
Parameter	**Parameter description**
error	Error message for loading status. If the loading is successful, the errorCode is FinAuthLiveV5DetectErrorTypeOK. Loading failed. Please check the specific reason for the failure. If the loading fails, the "FinAuthLiveV5EndDetectBlock" and "FinAuthLiveV5DismissBlock" will no longer be triggered
extraOutDict	Information return. Generally nil.

3.2.5.2 FinAuthLiveV5EndDetectBlock

Block Name	Block when the FinAuth liveness detection configuration interface finishes detecting
typedef void(^FinAuthLiveV5EndDetectBlock)(FinAuthLiveV5DetectError* error, NSString* bizTokenStr, NSDictionary* extraOutDict);	
Parameter	**Parameter description**
error	Error message when liveness detection ends. If the detection is successful, the errorCode is FinAuthLiveV5DetectErrorTypeOK. If the detection fails, please check the specific failure reason. Whether success or failure, will trigger "FinAuthLiveV5DismissBlock"
bizTokenStr	The business serial number passed in when initializing the SDK.
extraOutDict	Information return. If there is local retention information, user incoming information, etc., return through this parameter. When the model is FinAuthLiveV5DetectModel_C, the SDK output information necessary for verification will be returned in extraOutDict.

3.2.5.3 FinAuthLiveV5DismissBlock

Block Name	Block when the FinAuth liveness detection configuration interface page is collapsed and completed
typedef void(^FinAuthLiveV5DismissBlock)(void);	

Parameter	None

3.2.5.4 FinAuthLiveV5NetworkRequestBlock

Block Name	FinAuth liveness detection configuration interface manager network request status Block
typedef void(^FinAuthLiveV5NetworkRequestBlock)(int networkType, NSString* bizTokenStr, NSString* dataStr, FinAuthLiveV5NetworkResponseBlock configResponse);	
Remarks	This parameter is only valid when the model is FinAuthLiveV5DetectModel_E. The block is a sub-thread. If you need to perform UI operations, please return to the main thread manually.
Parameter	**Parameter description**
networkType	networkType is the type of the internal network request of the SDK. Please select different interfaces for network requests according to the network type. When networkType is 1, bizTokenStr, dataStr and configResponse are valid. Please use the datasync interface of the API and configure data_type to 1. When networkType is 2, bizTokenStr and dataStr are valid. Please use the datasync interface and configure data_type to 2.
bizTokenStr	Required parameters for the datasync interface
dataStr	Required parameters for the datasync interface
configResponse	Pass data to the SDK using blocks.

3.2.5.5 FinAuthLiveV5NetworkResponseBlock

Block Name	Block when the FinAuth liveness detection configuration interface page is collapsed and completed
typedef void(^FinAuthLiveV5NetworkResponseBlock)(NSDictionary* responseDict);	
Remarks	This parameter is only valid when the model is FinAuthLiveV5DetectModel_E.

	Call this block on the default thread or a non - main thread. Calling this block on the main thread will cause the network to be laggy until it times out.
Parameter	None
responseDict	Response body content of the network interface

3.3 HarmonyOS Interface Description

3.3.1 Entry class FaceLiveManager

The FaceLiveManager class is a class that manages liveness authentication. This class cannot be initialized and can only obtain its singleton through getInstance.

3.3.1.1 Get the singleton object of FaceLiveManager

Function name	getInstance	
public static getInstance(): FaceLiveManager		
Description	Get an instance of FaceLiveManager	
Type description	function	
Remarks	Get the singleton of FaceLiveManager	
Variable name	**Remarks**	**Type**
None		
Return value type	**Remarks**	
FaceLiveManager	An example	

3.3.1.2 Start Detection

Function name	startDetect
public startDetect(context: Context, config: FaceLiveDetectConfig, callback: FaceLiveDetectListener, uiContext?: UIContext)	
Description	Start liveness detection
Type description	function
Remarks	Open the liveness detection interface

Variable name	Remarks	Type
context	Context	Context
config	Parameter configuration, see FaceLiveDetectConfig for details	FaceLiveDetectConfig
callback	The implementation class of the FaceLiveDetectListener interface, which is used to receive the liveness detection results. For details, see the description of FaceLiveDetectListener.	FaceLiveDetectListener
uiContext	It can be passed or not. When passed, the page jump is processed through uiContext.getRouter()	UIContext
Return value type	**Remarks**	
void	None	

3.3.1.3 Get log information during liveness detection

Function name	getSDKLog	
public getSDKLog(): string;		
Description	Get the log information during the liveness detection process	
Type description	function	
Remarks	Get the log information during the liveness detection process	
Variable name	**Remarks**	**Type**
None		
Return value type	**Remarks**	
string	Encrypted log information during the liveness detection process	

3.3.1.4 Decrypt the video and image files during liveness detection

Function name	getLivenessFiles

public getLivenessFiles(filePath: string, decryptionKey: string): LivenessFileResult;		
Description	Decrypt locally stored files	
Type description	function	
Remarks	Decrypt locally stored files	
Variable name	**Remarks**	**Type**
filePath	The path of the encrypted liveness file returned by the SDK after the liveness is successfully uploaded	string
decryptionKey	Decryption key obtained after calling verify	string
Return value type	**Remarks**	
LivenessFileResult	See 3.3.3.5 for the decryption results of liveness files LivenessFileResult	

3.3.1.5 Get SDK version number information

Function name	getVersion	
public static getVersion(): string;		
Description	Get the SDK version number	
Type description	function	
Remarks	Get the SDK version number	
Variable name	**Remarks**	**Type**
None		
Return value type	**Remarks**	
string	The current SDK version number	

3.3.1.6 Get SDK build version information

Function name	getBuildInfo
public static getBuildInfo(): string;	

Description	Get the build information of the SDK	
Type description	function	
Remarks	Get the build information of the SDK	
Variable name	Remarks	Type
None		
Return value type	Remarks	
string	Build Information	

3.3.2 Detection Configuration Class

3.3.2.1 FaceLiveDetectConfig

Variable name	Type	Remarks
bizToken	string	Request token, must be passed
host	string	Service host Default to domestic
language	string	SDK language, the parameter is passed in according to the Language codes-ISO 639 standard
isShowLogo	boolean	Whether to display the bottom logo
autoAdjustVolume (reserved field)	boolean	[HarmonyOS is not supported] Whether to automatically adjust the volume of the active liveness voice prompt. Optional for compatibility mode. Default is false.
suggestVolume (reserved field)	boolean	[HarmonyOS is not supported for the time being] Adjust the recommended volume value. When the compatibility mode autoAdjustVolume is true, it is optional. The default value is 50
verticalDetection	FaceLiveVerticalDetectType	Vertical detection type, see 3.3.2.3 FaceLiveVerticalDetectType for details
uiConfigItem	FaceLiveDetectUIConfigItem | undefined	Check the UI configuration of the page. For details, see 3.3.2.4 FaceLiveDetectUIConfigItem
showV5Loading	boolean	

		Whether to display the loading page before starting the detection page
mode	FaceLiveDetectModel	Detection mode, see 3.3.2.2 FaceLiveDetectModel for details
configInfo	string	Configuration information (used in compatibility mode, data returned by biz_token interface)
standbyURLList	string[]	host list (used in compatibility mode. When the host request fails, the hosts in the list are requested cyclically)
uploadTimeout	number	Timeout setting for upload interface (unit: s)
tipsMode	TipsShowMode	Tips display mode. The default is MODE_SHORT. See 3.3.2.5 TipsShowMode for details
secureCamera	boolean	Whether to turn on the security camera. The default value is false, which means not to turn on
requestMaxTime	number	Pass-through mode config request waiting time, unit: s. Default is 30
isShowExitPopup	boolean	Whether to display the confirmation dialog when closing. Default: true
landscape	boolean	Whether it is landscape. Default: false
isMute	boolean	Whether to mute. Default: false
isHiddenActionTime	boolean	The action countdown is hidden during the action stage of Dynamic liveness true: Hidden false: Show

3.3.2.2 FaceLiveDetectModel

Type	Corresponding value	Remarks
MODEL_A	1	The default integration method, which authorizes and obtains the lliveness configuration through network

		requests inside the SDK, needs to be used in conjunction with the specified `hostURL` parameter.
MODEL_C	3	Compatibility integration mode, only authorization is performed through network requests inside the SDK, and the liveness configuration needs to be actively passed in by the user, which needs to be used in conjunction with the specified `hostURL` and `configInfo` parameters.
MODEL_D	4	V2 compatibility mode
MODEL_E	5	Transparent Mode

3.3.2.3 FaceLiveVerticalDetectType

Type	Corresponding value	Remarks
Continue	1	Enable device vertical detection
Front2	2	Only enabled within 2 seconds after detection is turned on, and then the function is turned off
Disable	3	Disable device vertical detection

3.3.2.4 FaceLiveDetectUIConfigItem

Variable name	**Type**	**Remarks**
livenessHomeNormalRemindTextColor	string	Hint text color in normal state, default color is #292929
livenessHomeProcessBarColor	string	The color of the progress bar in the liveness stage is only valid for the active liveness mode and the flash liveness without light mode. The default color is #267CE0
livenessHomeCheckingLineStartColor	string	The color of the progress bar during the verification phase. The default start color is rgb(38,124,224)
livenessHomeCheckingLineEndColor	string	Color of the progress bar during the verification phase. [Not supported on Hongmeng for the time being]
livenssHomeActionLineDoingColor	string	

		The line state color when the private cloud Dynamic liveness action phase is not completed. The default color is rgb(216,216,216)
livenessHomeActionLineDoneColor	string	The line state color when the private cloud Dynamic liveness action phase is completed.
livenessHomeExitPopupwindowTextSize	number	Exit Popup Title Font Size
livenessHomeExitPopupwindowBodySize	number	Exit the pop - up box body font size
livenessHomeConfirmButtonColor	string	Exit the color of the confirmation button in the pop - up box
livenessHomeCancelButtonColor	stirng	Exit the color of the cancel button in the pop - up box
livenessHomeAgreementpageTitleTextSize	number	Font size of the top title on the agreement page
livenessHomeAgreementpageBottomTitleTextSize	number	Font size of the prompt at the bottom of the agreement page
livenessHomeAgreementpageBottomButtonBeforeClickColor	string	Normal state of the button on the agreement page
livenessHomeAgreementpageBottomButtonAfterClickColor	string	Highlight state of the button on the agreement page
livenessHomeCustomPromptBackgroundColor	string	Liveness interface global prompt background color
livenessHomeCustomPromptTextColor	string	Global prompt font color for the liveness interface
livenessCustomPromptColor	string	Prompt font color for the liveness interface confirmation mode
livenessHomeBackgroundColor	string	Background color of non - face area
livenessHomeFailedRemindTextColor	string	Non-first mirror text color in the liveness interface

3.3.2.5 TipsShowMode

Name	TipsShowMode

Type description	Prompt mode type
Remarks	
Enumeration Name	**Remarks**
MODE_SHORT	Short mode
MODE_LONG	Long mode

3.3.3 Detection and monitoring callbacks

3.3.3.1 FaceLiveDetetListener

3.3.3.1.1 onPreDetectFinish

Function name	onPreDetectFinish	
onPreDetectFinish(errorCode: number, errorMessage: string): void;		
Description	Preprocessing result callback	
Type description	function	
Remarks	Preprocessing result callback	
Variable name	**Remarks**	**Type**
errorCode	See 3.3.4 for error codes	number
errorMessage	Error description	string
Return value type	**Remarks**	
void	None	

3.3.3.1.2 onDetectFinish

Function name	onDetectFinish
onDetectFinish(errorCode: number, errorMessage: string, bizToken: string, v2FaceLiveData: FaceLiveDetectDataStruct|undefined, v5FaceLiveData: FaceLiveResultData): void;	
Description	Callback for detection results
Type description	function

Remarks	Callback for detection results	
Variable name	**Remarks**	**Type**
errorCode	See 3.3.7 for error codes	number
errorMessage	Error description	string
bizToken	Business number	string
v2FaceLiveData	Only returned in v2 mode (config.mode = 4), returns undefined in normal mode	FaceLiveDetectDataStruct | undefined, see 3.3.3.2 FaceLiveDetectDataStruct for details
v5FaceLiveData	Return of test results	FaceLiveResultData | undefined, see 3.3.3.3 FaceLiveResultData for details
Return value type	**Remarks**	
void	None	

3.3.3.1.3 onLivenessLocalFileCallback

Function name	onLivenessLocalFileCallback	
onLivenessLocalFileCallback(livenessFilePath: string): void;		
Description	Callback for the result of the retention of liveness data, including the retention of liveness data	
Type description	function	
Remarks		
Variable name	**Remarks**	**Type**
fileInfo	Local return file, see FaceLiveLocalFileInfo for details	FaceLiveLocalFileInfo
Return value type	**Remarks**	
void	None	

3.3.3.1.4 onLivenessFileCallback

Function name	onLivenessFileCallback

onLivenessFileCallback(livenessFilePath: string): void;		
Description	Callback for the result of the retention of liveness data, including the retention of liveness data	
Type description	function	
Remarks		
Variable name	**Remarks**	**Type**
fileInfo	Local file path	string
Return value type	**Remarks**	
void	None	

3.3.3.2 FaceLiveDetectDataStruct

Only used in V2 mode

Name	V2 mode detection completed return data type	
FaceLiveDetectDataStruct		
Type description	Class	
Remarks		
Variable name	**Remarks**	**Type**
delta	delta	string
images	List of locally retained files	Map<stirng, ArrayBuffer>

3.3.3.3 FaceLiveDetectResultData

Only used in V5 compatibility mode

Name	V5 compatibility mode detection completion return data type	
FaceLiveDetectResultData		
Type description	Class	
Remarks		
Variable name	**Remarks**	**Type**

delta	delta	string
bestImage	BestImage returned in plain text	ArrayBuffer

3.3.3.4 FaceLiveLocalFileInfo

Name	Local retention file class	
FaceLiveLocalFileInfo		
Type description	Class	
Remarks		
Variable name	**Remarks**	**Type**
filePath	-	string
screenFilePath	Screen recording file path	string

3.3.3.5 LivenessFileResult

Name	Locally retained files	
LivenessFileResult		
Type description	Class	
Remarks		
Variable name	**Remarks**	**Type**
resultCode	Error Code	number
livenessType	Livenss Type	string
livenessFiles	List of locally retained files	LivenessFile[]， for details of LivenessFile, see 3.3.3.6 LivenessFile

3.3.3.6 LivenessFile

Name	Locally retained files
LivenessFile	

Type description	Class	
Remarks		
Variable name	**Remarks**	**Type**
path	File path	number
fileType	File type (Video: 1, Image: 2)	string
actionType	Active liveness type	string
isScreenRecord	Whether to record the screen	boolean
isBestImage	Is it the best image?	boolean

3.3.4 Error Code

ErrorCode	ErrorMessage	Remarks
1000	LIVENESS_FINISH	Liveness completion
	GET_CONFIG_SUCCESS	Successfully obtained the configuration
1001	BIZ_TOKEN_DENIED	The incoming biz_token does not meet the requirements
1002	ILLEGAL_PARAMETER:{livenesstype} ILLEGAL_PARAMETER:{Context} ILLEGAL_PARAMETER:{ config } ILLEGAL_PARAMETER:{config.mode} ILLEGAL_PARAMETER:{config.host} ILLEGAL_PARAMETER:{config:configInfo} ILLEGAL_PARAMETER:{config:isPrivate} ILLEGAL_PARAMETER:{config:bizToken} ILLEGAL_PARAMETER: {config:detectModelBuffer} ILLEGAL_PARAMETER:{config:isPrivate} ILLEGAL_PARAMETER:{missing_liveness_config} ILLEGAL_PARAMETER:{response_result_is_null} ILLEGAL_PARAMETER:{request_data_is_null}	The imported parameter is illegal, and the specific reason is in {}

	ILLEGAL_PARAMETER:{request_data_error}	
1003	AUTHENTICATION_FAIL:{illegal_param} AUTHENTICATION_FAIL:{illegal_handle} AUTHENTICATION_FAIL:{illegal_index} AUTHENTICATION_FAIL:{expire} AUTHENTICATION_FAIL:{bundleid_error} AUTHENTICATION_FAIL:{license_error} AUTHENTICATION_FAIL:{liveness_id_error} AUTHENTICATION_FAIL:{model_error} AUTHENTICATION_FAIL:{algo_error} AUTHENTICATION_FAIL:{opengl_context_error}	Authentication failed, the specific reason is in {}
1004	MOBILE_PHONE_NOT_SUPPORT	The phone is not supported
1005	Specific exception information	If such an error occurs, it is recommended to contact technical support in time
1006	REQUEST_FREQUENTLY	When the same device is called multiple times at the same time, the first call runs normally, and other calls return this type of error
1007	NETWORK_TIME_OUT	Network request timed out
1008	INTERNAL_ERROR	Network configuration error. When this error occurs, please request again. If this error persists, please contact customer service or business in time.
1009	INVALID_BUNDLE_ID	Information verification failed. Please restart the program or device and try again.
1010	NETWORK_ERROR	Unable to connect to the Internet. Please connect to the Internet and try again.
1011	USER_CANCEL	User cancelled
1012	NO_CAMERA_PERMISSION	You don't have permission to open the camera. Please enable the permission and try again.

1013	DEVICE_NOT_SUPPORT	Unable to start the camera. Please make sure the camera is working properly.
1014	FACE_INIT_FAIL	Unable to start facial recognition, please try again later
1016	LIVENESS_FAILURE	Liveness detection failed
1017	GO_TO_BACKGROUND	The app goes to the background and the liveness detection fails
1018	LIVENESS_TIME_OUT	Operation timed out because the user did not perform any operations for a long time
1019	DATA_UPLOAD_FAIL	Data upload failed
1022	MOBILE_PHONE_NOT_SUPPORT_SCRN	The device does not support screen recording
1023	SCRN_AUTHORIZATION_FAIL	Screen recording authorization failed
1024	SCRN_RECORD_FAIL	Video recording failed
1025	VIDEO_SAVE_FAIL	Video save failed
1026	NO_AUDIO_RECORD_PERMISSION	No permission to record voice
1027	SECURE_CAMERA_FAILED	Failed to initialize the security camera
1028	WINDOW_STATUS_FAIL	Window mode does not meet the requirements

4 SDK Instructions

4.1 Android Usage Instructions

4.1.1 Android UI customization

Customizable UI content includes:

1) Modify the size, color, and text content

Modify parameters such as text content, size, and color in the values folder.

2) Replace the image resource

Replace the image resources stored in the drawable folder. The resource key values need to be defined in accordance with the specified format.

3) Replace voice resources

Replace the audio file in the raw folder.

4.1.2 Android Resource External Loading

1. Store the model locally via network or other means, and pass the path of the model file to the SDK;
2. The resource file needs to be built into the main project res in advance.

4.1.3 Multilingual Support

1. Set the language (zh/en) and region through the language and region parameters in MegLiveDetectConfig.
2. The text language can be modified by replacing the value in the livenessv5_sdk_strings.xml file in the values folder corresponding to the language and region. For example, for English, modify the values-en folder, and for Chinese, modify the values-zh folder.
3. For voice broadcast, you need to replace the audio files in the corresponding raw folder. For example, for English, you need to modify the raw-en folder, and for Chinese, you generally need to modify the raw-zh folder.

4.2 iOS Usage Instructions

4.2.1 iOS UI customization

When using the SDK interface `+faceLiveDetectManagerWithBizToken:` for liveness authentication, set the **`customUI` of the `FinAuthLiveV5DetectInitConfigItem`** class to a custom **`FinAuthLiveV5DetectUIConfigItem`** object.

The properties of this object have default parameters. Please adjust the parameters according to your own needs. You can set the UI prompt display effect, UI element color, and UI element font size.

4.2.2 Multilingual Support

The iOS SDK supports user-defined configuration for multi-language support. The domestic version of the SDK comes with Chinese resources by default, while the overseas version comes with Chinese and English resources by default. For other language types, users need to add them by themselves.

This section uses French as an example to introduce the process of adding support for a language.

1. Modify the `FinAuthLiveV5DetectLanguageType` in **FaceLiveV5DetectConfig.h** and add the French type `FinAuthLiveV5DetectLanguageTypeFr = 2, // French`;
2. Add resources to the effect package **FinAuthLiveV5Detect.bundle** . Create a new language package folder named `fr` in each of the `audio`, `image`, and `text` resource folders. Then add the corresponding resources to the `fr` folder. For the voice package, add an m4a file with the same name. For the image package, add a png file with the same name. For the text package, create a new character file named `FinAuthFaceLiveV5.strings` and configure the specific language text content in this file.
3. Modify the `languageconfiguration.plist` in the effect package **FinAuthLiveV5Detect.bundle** and add a new item, say `item2`, to `FaceLiveV5Language`;
4. Set the type of `item2` to Dictionary and add three keys, `type`, `enum`, and `mask`.
   The value of `type` is of string type and is only used for annotation and does not have any actual functional purpose. It can be added at will. Here it is set to `FinAuthLiveV5DetectLanguageTypeFr`;
   The value of `enum` is of Number type and must be the same as the value in `FinAuthLiveV5DetectLanguageType`. Here it is set to `2`;
   The value of `mask` is of string type and must be the same as the name of the new language package folder in the effect package. Here it is set to `fr`;
5. When using the SDK interface `+faceLiveDetectManagerWithBizToken:` for liveness authentication, set the `languageType` of the **FinAuthLiveV5DetectInitConfigItem** class to the custom `FinAuthLiveV5DetectLanguageTypeFr` object;
6. After starting the SDK, the language of the UI is the configured French information.
7. If you only need to customize the voice and text, and do not need to customize the image, please ensure that there is a `ch` folder and the corresponding image resources in the `image`. After the SDK is started, if the image resources corresponding to the specified language cannot be found, the files in the `ch` folder will be used.

4.2.3 External resource loading

1. Add the **FinAuthLiveV5Detect.bundle** effect package to the sandbox by downloading it from the network or other means;
2. When using the SDK interface `+faceLiveDetectManagerWithBizToken:` for liveness authentication, set the `bundleFilePath` of the **FinAuthLiveV5DetectUIConfigItem** class to an absolute path, and the path should end with `bundle`;
3. If the file path is not specified or the specified path is abnormal, and no resources are added to the MainBundle of the project, when the SDK is started for liveness authentication, the `NO_ADD_RESOURCE` error will be returned.

4.3 HarmonyOS Usage Instructions

4.3.1 HarmonyOS UI customization

Set the FaceLiveDetectUIConfigItem object through the uiConfigItem in FaceLiveDetectConfig. The attributes of this object have default parameters. Please adjust the parameters you need to adjust by yourself. You can set the UI prompt display effect, the color of UI elements and the font size of UI elements.

4.3.2 Multilingual Support

1. Set the voice (zh/en) through the language in FaceLiveDetectConfig.
2. The text voice modifies the display content by replacing the value in resource/string.json
3. For voice broadcast, the voice files need to be placed in resource/rawfile, and the file names cannot be modified. If it is English, add the suffix _en and configure the corresponding voice file name in resource/string.json.

5 Resource list

5.1 Voice Resource List

Voice Name (IOS&Android)	Voice Name (**Harmony**)	Voice content	Remarks
liveness_bllink.m4a	liveness_bllink.m4a	Please blink	The blinking action is valid for both active liveness and dynamic liveness when multiple actions are performed
liveness_nod.m4a	liveness_nod.m4a	Please nod up and down	The nodding action is valid for both active liveness and dynamic liveness when multiple actions are required
liveness_mouth.m4a	liveness_mouth_open.m4a	Please open your mouth	The mouth opening action is valid for both active liveness and dynamic liveness have multiple actions
liveness_shakehead.m4a	liveness_shakehead.m4a	Please shake your head left and right	The shake head is valid for both active liveness and dynamic

			liveness when there are multiple actions
liveness_well_done.m4a	liveness_well_done.m4a	Very good	Active Liveness and Dynamic liveness are valid when combining multiple actions
liveness_action_confirm	liveness_action_confirm	Empty by default No such file	The audio corresponding to the prompt text of the Dynamic liveness confirmation mode needs to be configured by the customer according to the prompt text configured by themselves
liveness_move_back.m4a	liveness_move_back.m4a	Move Back a Little	Distance Liveness is valid
liveness_move_forward.m4a	liveness_move_forward.m4a	Move Forward a Little	Distance Liveness is valid
liveness_approach_slowly.m4a	liveness_approach_slowly.m4a	Slowly move phone close to face	Distance Liveness is valid
liveness_hold_still.m4a	liveness_hold_still.m4a	Hold Still for a Moment	Distance Liveness is valid

5.2 Image Resource List

IOS image name	Android image name	**Harmony image name**	Image content	Remarks
liveness_action_ai_nod	liveness_action_nod_tips_icon	liveness_action_ai_nod	Nodding up and down action guidance	Display when the remaining time of the active liveness nodding action is less than 50%
liveness_action_ai_shake	liveness_action_shake_tips_icon	liveness_action_ai_shake	Left and right head turning action guidance	Display when the remaining time of the active liveness head turning

				action is less than 50%
liveness_action_down	liveness_action_down	liveness_action_down	The picture of the head down in the nodding animation	Active liveness is valid
liveness_action_eye_close	liveness_action_eye_close	liveness_action_eye_close	Closed eye image in blink animation	Active liveness is valid
liveness_action_left	liveness_action_left	liveness_action_left	Left turn picture in turn animation	Active liveness is valid
liveness_action_right	liveness_action_right	liveness_action_right	The picture of turning the head to the right in the turning head animation	Active liveness is valid
liveness_action_mouth_open	liveness_action_mouth_open	liveness_action_mouth_open	Mouth opening picture in mouth opening animation	Active liveness is valid
liveness_action_mouth_close	liveness_action_mouth_close	liveness_action_mouth_close	Closed mouth image in mouth opening animation	Active liveness is valid
liveness_action_normal	liveness_action_normal	liveness_action_normal	Picture of the normal state of the active liveness	Active liveness is valid
liveness_action_success_icon@2x	liveness_action_success_icon	liveness_action_success_icon	Pictures passed during the action phase of Dynamic liveness	Dynamic liveness confirmation mode is effective.

liveness_distance_eye_frame		liveness_flash_eye_frame	Guide eye alignment picture after the Dynamic liveness offset function is enabled	Distance Liveness is valid
liveness_distance_mirror_light@2x	liveness_distance_mirror_light	progress_dot	Distance from the liveness to the mirror stage animation guide picture	Distance Liveness is valid
liveness_distance_move_image1@2x	liveness_distance_move_image1	progress_close_to_1	Animation guide picture for the stage when the Dynamic liveness moves closer again	Distance Liveness is valid
liveness_distance_move_image2@2x	liveness_distance_move_image2	progress_close_to_2	Animation guide picture for the stage when the Dynamic liveness moves closer again	Distance Liveness is valid
liveness_flash_eye_frame		liveness_flash_eye_frame	Guide eye alignment picture after the flash liveness enables the offset function	Flash liveness is valid
liveness_home_action_line_icon1@2x	liveness_home_action_line_icon1	liveness_home_action_line_icon1	Dynamic liveness multi-action stage prompt animation	Dynamic liveness multi-action effective
liveness_home_action_line_icon2@2x	liveness_home_action_line_icon2	liveness_home_action_line_icon2	Dynamic liveness multi-	Dynamic liveness multi-

			action stage prompt animation	action effective
liveness_home_action_line_icon3@2x	liveness_home_action_line_icon3	liveness_home_action_line_icon3	Dynamic liveness multi-action stage prompt animation	Dynamic liveness multi-action effective
liveness_home_action_line_icon4@2x	liveness_home_action_line_icon4	liveness_home_action_line_icon4	Dynamic liveness multi-action stage prompt animation	Dynamic liveness multi-action effective
liveness_home_agreement	liveness_home_agreement	liveness_home_agreement	Protocol page prompt picture	
liveness_agreement_noselected@2x	liveness_agreement_noselected	agreement_unclicked	Picture of unchecked checkbox on the agreement page	
liveness_agreement_selected@2x	liveness_agreement_selected	agreement_clicked	Check box selected image on the agreement page	
liveness_home_back_highlight@2x	liveness_home_back_highlight		Highlight state of the return button on the agreement page	
liveness_home_back_normal@2x	liveness_home_back_normal	icon_back_blue	The normal state of the return button on the agreement page	
liveness_home_close	liveness_home_close	liveness_home_close	Liveness page close button	Display during liveness

liveness_home_white_close	liveness_home_white_close	liveness_home_close_white	Liveness page close button	Effective during the flash liveness lighting stage
liveness_home_mask_contour		liveness_home_mask_contour	Face frame guided picture in the liveness stage	
liveness_home_mask		liveness_home_mask	Face frame guided picture in the liveness stage	
liveness_home_vertical_remind	liveness_home_vertical_remind		Device vertical detection prompt picture	Effective when the vertical detection phase gives an abnormal prompt
liveness_loading_icon@2x	liveness_loading_icon	liveness_loading_icon	Turn on the picture of the SDK Loading phase during the initialization process	It is valid after the initialization Loading is started
liveness_logo_icon	liveness_logo_icon	powered_by	Bottom logo image	

5.3 Text Resource List

IOS text key	**Android text key**	**Harmony text key**	Text content	Remarks
livenessHomePromptBlinkText	livenessV5HomePromptBlinkText	livenessV5HomePromptBlinkText	Please blink	Active liveness blink action prompt
livenessHomePromptOpenMouthText	livenessV5HomePromptOpenMouthText	livenessV5HomePromptOpenMouthText	Please open your mouth	Active liveness open mouth action prompt

livenessHomePromptShakeHeadText	livenessV5HomePromptShakeHeadText	livenessV5HomePromptShakeHeadText	Please shake your head left and right	Active liveness head turning action prompt
livenessHomePromptNodText	livenessV5HomePromptNodText	livenessV5HomePromptNodText	Please nod up and down	Active liveness nod action prompt
livenessHomePromptFrontalFaceText	livenessV5HomePromptFrontalFaceText	livenessV5HomePromptNoFaceText	Please align your face to the frame	Mirror stage prompt
livenessHomePromptFrontalEyeText	livenessV5HomePromptFrontalEyeText	livenessLookMirrorQualityFaceOutOfRectOffsetOpen	Please align your eyes with the guide box	Mirror stage prompt after enabling the offset function
livenessHomePromptDarkerText	livenessV5HomePromptDarkerText	livenessLookMirrorQualityErrorFaceTooBright	Please increase the lighting	Mirror stage prompt
livenessHomePromptBrighterText	livenessV5HomePromptBrighterText	livenessLookMirrorQualityErrorFaceTooDark	Please decrease the lighting	Mirror stage prompt
livenessHomePromptCloserText	livenessV5HomePromptCloserText	livenessLookMirrorQualityErrorFaceTooSmal	Please move closer to the camera	Mirror stage prompt
livenessHomePromptFurtherText	livenessV5HomePromptFurtherText	livenessLookMirrorQualityErrorFaceTooLarge	Please move farther away from the camera	Mirror stage prompt
livenessHomePromptNoEyesOcclusionText	livenessV5HomePromptNoEyesOcclusionText	livenessLookMirrorQualityOcclusionEye	Do not cover your eyes	Mirror stage prompt
livenessHomePromptNoMouthOcclusionText	livenessV5HomePromptNoMouthOcclusionText	livenessLookMirrorQualityOcclusionMouth	Do not cover your mouth	Mirror stage prompt
livenessHomePromptStayStillText	livenessV5HomePromptStayStillText	livenessLookMirrorQualityKeeping	Good. Hold still	Mirror stage prompt
livenessHomePromptAngleOffsetText	livenessV5HomePromptAngleOffsetText	livenessLookMirrorQualityErrorFaceRollDeviated	Please adjust the angle of your head	Mirror phase prompt

livenessHomePromptMotionBlurText	livenessV5HomePromptMotionBlurText		Please keep the shooting clear	Mirror stage prompt
livenessHomePromptVerticalText	livenessV5HomePromptVerticalText	livenessLookMirrorQualityNotVertical	Please keep your phone vertical	Mirror phase prompt
livenessHomeExitPopupwindowTitleText	livenessV5HomeExitPopupwindowTitleText	livenessHomeExitPopupwindowTitleText	Prompt	Pop - up title after clicking the close button
livenessHomeExitPopupwindowBodyText	livenessV5HomeExitPopupwindowBodyText	livenessHomeExitPopupwindowBodyText	Exit the verification process?	Pop - up content after clicking the close button
livenessHomeExitPopupwindowSureText	livenessV5HomeExitPopupwindowSureText	livenessHomeExitPopupwindowSureText	Confirm	Popup exit button content after clicking the close button
livenessHomeExitPopupwindowCancelText	livenessV5HomeExitPopupwindowCancelText	livenessHomeExitPopupwindowCancelText	Cancel	Popup retry button content after clicking the close button
livenessHomePromptCheckMultiFaceText	livenessV5HomePromptCheckMultiFaceText	livenessLookMirrorQualityMultiFace	Please ensure individual verification	Mirror phase prompt
livenessHomePromptEnvDarkerText	livenessV5HomePromptEnvDarkerText	livenessLookMirrorQualityEnvTooDark	The light is too dark. Please go to a brighter environment	Mirror phase prompt
livenessHomePromptEnvBrighterText	livenessV5HomePromptEnvBrighterText	livenessLookMirrorQualityEnvTooBright	The light is too bright. Please go to a darker environment	Mirror stage prompt
livenessHomePromptFaceEreaText	livenessV5HomePromptFaceEreaText	livenessLookMirrorQualityErrorFaceNonIntegrity	The effective face are is too small	Mirror stage prompt
livenessHomePromptWaitText	livenessV5HomePromptWaitText	livenessVerifying	Verification in progress...	Prompt for the completion stage

			Please wait	of the liveness process
livenessHomeAgreementText	livenessV5HomeAgreementText	livenessHomeAgreementText	User Agreement	User Agreement Link Title
livenessHomeCreditText	livenessV5HomeCreditText	livenessHomeCreditText	Notification on Personal Credit Information	User Information Notice Link Title
livenessHomeAgreementpageTopTitleText	livenessV5HomeAgreementpageTopTitleText	livenessHomeAgreementpageTopTitleText	Liveness Verification	User Agreement Page Title
livenessHomeAgreementpageBottomTitleText	livenessV5HomeAgreementpageBottomTitleText	livenessHomeAgreementpageBottomTitleText	Please face the screen directly and do not hide your face	User Agreement page prompt content
livenessHomeAgreementpageBottomButtonText	livenessV5HomeAgreementpageBottomButtonText	livenessHomeAgreementpageBottomButtonText	Start	User Agreement Page Start Button Content
livenessHomeScrnAuthorizedRejectText	livenessV5HomeScrnAuthorizedRejectText		This business requires granting screen recording permission. Please re-operate and grant permission or contact the platform customer service staff	Pop - up prompt content for enabling the screen recording function
livenessHomeScrnAuthorizedRejectButtonText	livenessV5HomeScrnAuthorizedRejectButtonText		Confirm exit?	Pop-up button content for enabling the screen recording function

livenessHomeCloserPromptText	livenessV5HomeCloserPromptText	livenessLookMirrorDistanceMoveClosely	Please slowly approach the screen and align the face frame again	Distance Liveness approaching stage prompt
livenessHomeCloserPromptImageText	livenessV5HomeCloserPromptImageText	livenessLookMirrorQualityErrorFaceTooSmal	Please hold your phone closer	Distance Liveness approaching stage prompt
livenessHomeLoadingpageText	livenessV5HomeLoadingpageText	livenessHomeLoadingpageText	Configuration is being loaded, please wait	Prompt for the Loading phase of SDK initialization
livenessHomePromptCorrectActionText	livenessV5HomePromptCorrectActionText	livenessV5HomePromptCorrectActionText	Inconsistent with the prompt action	Prompt content when retrying the active liveness start error
livenessHomePromptCoherenceText	livenessV5HomePromptCoherenceText	livenessV5HomePromptCoherenceText	Discontinuous aciton	Prompt content when retrying the active liveness start error
livenessHomePromptFaceLocationText	livenessV5HomePromptFaceLocationText	livenessV5HomePromptFaceLostText	Please keep your face within the face frame	Prompt content when retrying the active liveness start error
livenessHomePromptActionSpeedText	livenessV5HomePromptActionSpeedText	livenessV5HomePromptActionSpeedText	The action speed is too fast	Prompt content when retrying the active liveness start error
livenessHomePromptMaskText	livenessV5HomePromptMaskText	livenessV5HomePromptMaskText	Please take off your mask	Prompt content when retrying the active liveness start error
livenessHomePromptFaceLostText	livenessV5HomePromptFaceLostText	livenessV5HomePromptFaceLostText	Please keep your face within the face frame	Prompt content when retrying the active liveness start error

livenessHomePromptNoFaceText	livenessV5HomePromptNoFaceText	livenessV5HomePromptFaceLocationText	Please keep your face within the face frame	Prompt content when retrying the active liveness start error
livenessHomePromptOcclusionText	livenessV5HomePromptOcclusionText	livenessV5HomePromptOcclusionText	Please avoid covering your face	Prompt content when retrying the active liveness start error
livenessHomeDynamicActionTimeOutText	livenessV5HomeDynamicActionTimeOutText	livenessHomeDynamicActionTimeOutText	Action timed out, please try again	Prompt content when retrying the active liveness start error
livenessHomeDynamicActionWrongText	livenessV5HomeDynamicActionWrongText	livenessHomeDynamicActionWrongText	Wrong action, please try again	Prompt content when retrying the active liveness start error
livenessHomeDynamicBlinkText	livenessV5HomeDynamicBlinkText	livenessV5HomeDynamicBlinkText	Please blink	Blinking action is effective when the lliveness is moving
livenessHomeDynamicOpenMouthText	livenessV5HomeDynamicOpenMouthText	livenessV5HomeDynamicOpenMouthText	Please open your mouth	The mouth opening action is effective when the dynamic liveness makes multiple actions
livenessHomeDynamicShakeHeadText	livenessV5HomeDynamicShakeHeadText	livenessV5HomeDynamicShakeHeadText	Please slowly turn your head left or right	The head-turning action is effective when the dynamic liveness makes multiple actions
livenessHomeDynamicNodText	livenessV5HomeDynamicNodText	livenessV5HomeDynamicNodText	Please nod your head slowly	Nodding action is valid when there are multiple actions in the dynamic liveness
	livenessV5HomeDynamicActionBlinkTex	livenessV5HomeDynamicActionBlin		

	t	kText		
	livenessV5HomeDynamicActionOpenMouthText	livenessV5HomeDynamicActionOpenMouthText		
	livenessV5HomeDynamicActionShakeHeadText	livenessV5HomeDynamicActionShakeHeadText		
	livenessV5HomeDynamicActionNodText	livenessV5HomeDynamicActionNodText		
	livenessV5HomeDynamicActionPromptText	livenessV5HomeDynamicActionPromptText		
	livenessV5HomeDynamicFlashPromptText	livenessV5HomeDynamicFlashPromptText		
	livenessV5HomeDynamicPromptModeText	livenessV5HomeDynamicPromptModeText		
	livenessV5HomeFlashPromptModeText	livenessV5HomeFlashPromptModeText		
	livenessV5HomeActivePromptModeText	livenessV5HomeActivePromptModeText		
	livenessV5HomeDistancePromptModeText	livenessV5HomeDistancePromptModeText		

5.4 UI Configurable Items

Android Parameter Name	Parameter Meaning	Default Configuration Value
livenessV5HomeProcessBarColor	Liveness Recording Progress Bar Color (Active liveness, flash liveness	#267CE0

	without lighting, Loading)	
livenessV5HomeDeviceVerticalRemindColor	Short mode vertical prompt font color	#FFFFFF
livenessV5HomeDeviceVerticalLongModeRemindColor	Long mode vertical prompt font color	#292929
livenessV5HomeNormalRemindTextColor	The color of the prompt text when entering the mirror stage for the first time	#292929
livenessV5HomeFailedRemindTextColor	Non - first mirror text color	#FF3B42
livenessV5HomeAgreementpageBottomButtonBeforeClickColor	Color of the previous button, protocol color, and return color before clicking	#3B94FC
livenessV5HomeAgreementpageBottomButtonAfterClickColor	Button color, protocol color, and return color after clicking	#267CE0
livenessV5HomeCustomPromptBackgroundColor	Customize the background color of business prompts	#FDEBEB
livenessV5HomeCustomPromptTextColor	Customize the font color of business prompts	#DA3130
livenessCustomPromptColor	Customize the icon color corresponding to the business prompt content	#FF0000
livenessV5HomeActionHatColor	Color of the small hat in the active liveness recording	#0F3159
livenessV5HomeNormalRemindTextColor	Color of the prompt text for flash liveness without lighting	#000000
livenessV5ExitRightPromptColor	Confirm button color	#007AFF
livenessV5ExitLeftPromptColor	Cancel button color	#777777
livenessV5HomeBackgroundColor	Background color of non - face area (including the top battery bar area)	#FFFFFF
livenessV5HomeFinishContourLineColor	Face frame color after liveness recording	#267CE0
livenessHomeActionLineDoingColor	Define the color of the default unfinished line in the multi - action	#D8D8D8

	of the Dynamic liveness (only effective for privatization)	
livenessHomeActionLineDoneColor	Define the color of the default completion line in the multi-action of the Dynamic liveness (only effective for privatization)	#267CE0

5.5 SDK Compatibility Mode Image List

liveness modes	Image name	Image meaning	Remarks
Active Liveness	image_best	Best face image	
	image_mirror	Best image in the mirror stage	
	image_nod	Nod action image in the action stage	Based on the configured number and type of actions, return the corresponding action images.
	image_shake	Shake head action image in the action stage	
	image_mouth	Open mouth action image in the action stage	
	image_blink	Blink action image in the action stage	
Flash Liveness	image_best	Best face image	
	image_flash1	Flash liveness lighting stage image 1	
	image_flash2	Flash liveness lighting stage image 2	
	image_flash3	Flash liveness lighting stage image 3	
Distance Liveness	image_best	Best face image	
	image_near_mirror	Best Image in Close-Up Mirror Stage	

	image_far_mirror	Best Image in Long-Distance Mirror Stage	
	image_flash1	Flash stage image 1	
	image_flash2	Flash stage image 2	
	image_flash3	Flash stage image 3	
Dynamic liveness	image_best	Best face image	
	image_nod	Nod action image in the action stage	Based on the configured number and type of actions, return the corresponding action images.
	image_shake	Shake head action image in the action stage	
	image_mouth	Open mouth action image in the action stage	
	image_blink	Blink action image in the action stage	
	image_flash1	Flash stage image 1	
	image_flash2	Flash stage image 2	
	image_flash3	Flash stage image 3	